**End Semestral Exam**
**Representation Theory of Finite Groups**
**BMath-IIIrd year**

---

Time: 3 hours

Answer as many questions as you can. Max marks 50. No marks will be awarded in absence of proper justification.

*Notations are as used in class. All groups are finite, and all representations are finite dimensional over* $\mathbb{C}$.

1. Show that the character table of a group $G$ is invertible when viewed as a matrix. (4)

2. Let $D_8$ denote the dihedral group of order 8.
(a) Show that there is only one irreducible representation of $D_8$ of dimension at least 2, say $V$.
(b) Find the character table of $D_8$.
(c) Find the decomposition of the tensor product $V \otimes V$ and the exterior product $\wedge^2(V)$, into irreducibles. (1+6+6)

3. (a) Let $H$ be a subgroup of a group $G$. Show that for every irreducible representation $(\rho, V)$ of $G$ there is an irreducible representation $(\rho', W)$ of $H$ such that $\rho$ is an irreducible component of $\mathrm{Ind}_H^G W$.
(b) Let $G$ be an abelian group, and $H$ be a subgroup. Let $\phi$ be an irreducible character of $H$. Show that there are exactly $[G : H]$ many irreducible characters $\psi$ of $G$, such that $\phi$ is the restriction of $\psi$ to $H$. (6+6)

4. (a) Prove that the number of real irreducible characters of $G$ equals the number of real conjugacy classes of $G$.
(b) Deduce that $G$ is of odd order if and only if $G$ does not have any non-trivial real irreducible characters. (8+6)

5. (a) Define Specht representation $S^\lambda$ of the symmetric group $S_n$ corresponding to the partition $\lambda$ of $n$.
(b) Prove that if $\lambda$ is the partition $(n-1, 1)$ of $n$, then the Specht representation $S^\lambda$ is equivalent to the standard representation of $S_n$. (4+8)

*************************